

Balancing Predictive Performance and Explanation Reliability in Imbalanced Loan Default Prediction

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Abstract. Loan default prediction is a high-impact classification problem in which predictive discrimination, minority-class sensitivity, and explanation quality must be considered jointly. This application study evaluates an end-to-end machine-learning and explainable-AI pipeline on a public mortgage lending dataset containing 148,670 records. Logistic regression and a decision tree are used as transparent baselines, while random forest and XGBoost represent higher-capacity ensemble models. The data are stratified into training, validation, and independent test partitions; class imbalance is addressed using training-only SMOTE, and the ensemble models are tuned through five-fold cross-validation. On the test set, XGBoost achieves an F1-score of 0.859, recall of 0.901, ROC-AUC of 0.983, and PR-AUC of 0.951. Random forest produces comparable discrimination and slightly higher accuracy, but lower recall. SHAP and LIME are then examined as complementary explanation mechanisms. Global SHAP analysis identifies interest rate and encoded credit and loan characteristics as the dominant model drivers. In a case-level explanation diagnostic, TreeSHAP is faster and substantially more stable and faithful than LIME, while LIME offers a compact, stakeholder-friendly local narrative. The study’s central practical finding is that model selection and explanation selection should be treated as a coupled governance decision: an accurate classifier without reliable explanations is insufficient, while an intuitive explanation that poorly reflects the underlying model can be misleading.

Keywords: Loan default prediction · Credit risk · Imbalanced classification · XGBoost · Explainable artificial intelligence · SHAP · LIME

1 Introduction

Credit decisions require more than a model that is correct on average. A lender must identify a sufficiently large proportion of borrowers who are likely to default, control the cost of false alarms, and provide explanations that are intelligible enough to support review and challenge. These requirements are difficult to satisfy simultaneously. Default events are commonly less frequent than non-default events, so accuracy can appear strong even when a classifier misses many

risky borrowers. Moreover, the tree ensembles that often improve predictive performance are substantially less transparent than traditional scorecard-style models.

Credit scoring has a long history of statistical modelling, with logistic regression and related methods valued for their relative simplicity and auditability [10]. Comparative studies nevertheless show that modern machine-learning algorithms frequently improve discrimination by capturing nonlinear effects and interactions that are difficult to pre-specify [12]. Random forest [4] and gradient-boosted trees such as XGBoost [7] are therefore attractive for operational risk screening. Their use, however, transfers part of the modelling problem from prediction to explanation.

Post-hoc explainable-AI methods are commonly proposed as a bridge between predictive power and transparency. LIME approximates a model locally with a sparse surrogate [15], whereas SHAP decomposes a prediction into additive feature contributions derived from Shapley values [13]. Both methods can produce useful narratives, but they answer different questions and have different failure modes. A visually convincing explanation is not necessarily stable across repeated runs, and a locally faithful explanation does not establish causality. These distinctions matter in lending, where correlated predictors, one-hot encoded categories, and imbalanced outcomes can make feature attributions fragile [9, 8, 1].

This paper reports an application-oriented study of loan default prediction that evaluates the predictive and explanatory layers together. The work is intentionally not presented as a new learning algorithm. Its contribution is an integrated empirical assessment and a set of transferable implementation lessons for practitioners developing explainable risk classifiers. Specifically, the paper:

- benchmarks two interpretable baselines and two ensemble methods using metrics that expose minority-class performance, including recall, F1-score, ROC-AUC, and PR-AUC;
- combines global and local SHAP analyses with a LIME explanation for the same selected case, and reports case-level diagnostics of runtime, stability, fidelity, and explanation compactness; and
- derives practical guidance on choosing a prediction model and an explanation method as a single deployment decision rather than as independent technical choices.

The study is positioned in the application stream because the primary value lies in the complete workflow, the observed trade-offs, and the limitations encountered when standard AI components are assembled for a realistic financial classification task.

2 Related Work

2.1 Credit Risk Modelling and Imbalanced Classification

Traditional credit-scoring models are designed around interpretable relationships between borrower characteristics and the probability of an adverse outcome.

Logistic regression remains a useful benchmark because coefficient signs and magnitudes can be inspected, transformed into scorecards, and challenged by domain specialists. Decision trees provide a second interpretable family by expressing decisions as hierarchical rules. Their weakness is that a single tree can be unstable and prone to overfitting.

Ensemble learning addresses these limitations by aggregating multiple weak or unstable learners. Random forest reduces variance through bootstrap aggregation and random feature selection [4]. XGBoost constructs trees sequentially, optimising a regularised objective that allows later learners to focus on residual errors [7]. In credit-risk applications, such models have been shown to capture nonlinear relationships and interactions more effectively than linear baselines [12, 14, 2]. The improvement is not automatic: performance depends on preprocessing, validation design, hyperparameter selection, and the metric used for optimisation.

Class imbalance is especially important. When the non-default class dominates, a classifier can obtain high accuracy by favouring the majority class. Recall is then useful because it measures the proportion of actual defaults detected, while precision indicates how many predicted defaults are correct. Their harmonic mean, the F1-score, provides a single summary when both error types matter. ROC-AUC evaluates ranking quality over thresholds, but it can remain optimistic when the negative class is much larger. PR-AUC is therefore reported as a complementary measure because precision-recall analysis is more sensitive to minority-class performance [17].

Data-level rebalancing is one way to improve learning from a minority class. SMOTE creates synthetic minority examples by interpolating between neighbouring observations [6]. Its placement in the workflow is critical: oversampling must be restricted to the training data, and ideally to each training fold during cross-validation, because applying it before partitioning can leak synthetic information into validation or test samples.

2.2 Explainability in Credit Decisions

The interpretability literature distinguishes between inherently interpretable models and post-hoc explanations of black-box models [3]. Rudin [16] argues that high-stakes decisions should prefer transparent models whenever an interpretable model can achieve comparable performance. In other settings, however, the predictive advantage of an ensemble may be sufficiently material that post-hoc explanations become part of the governance architecture. Credit risk is one such domain, and several studies have proposed SHAP-based scorecards or explanation layers for machine-learning systems [5, 11].

For a fitted model f and an observation x , SHAP represents the prediction as

$$f(x) = \mathbb{E}[f(X)] + \sum_{j=1}^p \phi_j, \quad (1)$$

where ϕ_j is the contribution assigned to feature j . TreeSHAP exploits the structure of tree ensembles to calculate these additive contributions efficiently. The resulting

Table 1. Positioning of the present application study.

Focus	Typical strength	Typical omission	Present study
Predictive bench- marking	Broad comparison of classifiers	Explanations pending validation	ap- Reports minority- without sensitive test metrics and a separate valida- tion check
XAI case study	Rich visual interpreta- tion	Limited comparison with model perfor- mance	Links explanations to the selected operating model
Explanation eval- uation	Stability or fidelity analysis	Limited application context	Reports case-level run- time, stability, fidelity, and compactness to- gether

values support global summaries, such as mean absolute attribution, and local explanations, such as force or waterfall plots.

LIME instead seeks an interpretable surrogate g in the neighbourhood of x :

$$\arg \min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g), \quad (2)$$

where π_x weights perturbed samples by proximity to x , $\mathcal{L}$ measures local disagreement with the original model, and $\Omega(g)$ penalises complexity. LIME can produce concise explanations, but the random perturbation process and the choice of neighbourhood may lead to instability. SHAP is more theoretically constrained, yet its attributions can still be affected by correlated features and the background distribution used for the explanation [1].

Existing credit-risk studies often emphasise either predictive performance or the visual interpretation of selected outputs. Gramegna and Giudici [9] compare the discriminative value of SHAP and LIME in credit risk, while Chen et al. [8] show that imbalance can affect interpretability as well as classification. The present work builds on this literature by placing prediction metrics and explanation diagnostics in one application workflow. Table 1 summarises the intended positioning.

3 Study Design

3.1 Data and Prediction Task

The study uses the public Loan Default Dataset distributed through Kaggle [18]. It contains 148,670 anonymised mortgage-related records and 34 original fields describing borrower, loan, property, and credit characteristics. The binary target, **Status**, distinguishes default from non-default. The original distribution is imbalanced, with roughly three non-default observations for each default observation.

The feature set contains both numerical and categorical variables. Examples include loan amount, property value, income, credit score, loan-to-value ratio, interest rate, loan purpose, credit type, business or commercial purpose, occupancy, age band, and co-applicant information. Identifiers and the observation year were removed because they were not intended to represent borrower risk. Variables with extensive missingness were excluded, and the interest-rate spread variable was removed because of redundancy with the interest rate.

The task is formulated as supervised binary classification. The objective is not merely to maximise overall accuracy, but to identify defaults while preserving useful precision and ranking performance. The workflow proceeds from data preparation and stratified partitioning to predictive evaluation and explanation audit.

3.2 Preprocessing and Partitioning

Numerical missing values were imputed with the median, reducing sensitivity to extreme values. Categorical missing values were replaced with the mode. Categorical predictors were converted by one-hot encoding, with one reference category dropped where appropriate to reduce exact multicollinearity for logistic regression.

The prepared data were divided using stratified sampling into 70% training, 15% validation, and 15% independent test partitions. The validation set was retained as an additional check after model selection; the test set was not used during tuning. SMOTE was applied only to the training partition. Numerical standardisation was used for logistic regression, with the transformation learned from training data and applied to validation and test data. Tree-based models were trained on the encoded, resampled data without relying on scaling.

This design separates three distinct purposes. The training data support parameter estimation and cross-validation. The validation data provide a check that the ranking observed during tuning is not an artefact of a particular fold configuration. The test data provide the final comparison reported in Section 4.

3.3 Models and Model Selection

Four models were selected to span an interpretability-performance continuum:

Logistic regression.

A linear probabilistic baseline with directly inspectable coefficients.

Decision tree.

A nonlinear but rule-based baseline that can represent threshold effects.

Random forest.

A bagged tree ensemble designed to reduce the variance of individual trees.

XGBoost. A regularised gradient-boosting implementation designed for high predictive performance on structured data.

Random forest and XGBoost hyperparameters were selected through grid search with five-fold cross-validation on the training data. The validation check produced F1-scores of 0.864 for random forest and 0.861 for XGBoost. XGBoost had the higher validation recall (0.902), indicating a more favourable sensitivity to default cases even though the F1-scores were close. The final choice was therefore based on the combination of validation behaviour and independent test performance rather than a single metric.

3.4 Predictive Evaluation

For the positive default class, precision, recall, and F1-score are defined as

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (3)$$

$$F_1 = 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (4)$$

Accuracy, ROC-AUC, and PR-AUC are also reported. Recall has particular operational relevance because a false negative corresponds to an actual default that is not flagged. Precision remains important because excessive false positives can unnecessarily restrict lending or increase manual review workload. No universal metric determines the best model; the appropriate balance depends on the institution’s relative costs and decision threshold.

3.5 Explanation Methods and Diagnostics

The explanation analysis focuses on the final XGBoost model. SHAP is used in three ways: a summary plot to show the distribution and direction of feature effects, a mean absolute SHAP plot to rank global importance, and local plots to decompose a selected prediction. A dependence plot is used to inspect the relationship between the interest rate and its SHAP contribution.

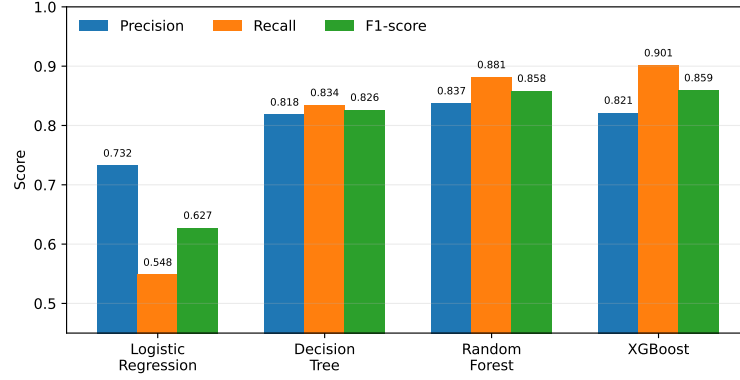
LIME is applied to the same selected test observation to create a sparse local surrogate. The comparison reports four implementation diagnostics from the original experiment:

- runtime for generating the selected explanation;
- a normalised stability score from repeated explanation generation;
- a local fidelity score measuring agreement with the underlying model around the selected observation; and
- the number of features required to account for approximately 80% of the reported local contribution.

These figures should be interpreted as a case-level diagnostic rather than a population-wide proof that one explanation method is universally superior. The diagnostic is useful because it exposes operational properties that are often omitted when SHAP or LIME plots are presented only as illustrations.

Table 2. Independent test performance for default as the positive class.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic regression	0.839	0.732	0.548	0.627	0.798	0.723
Decision tree	0.913	0.818	0.834	0.826	0.886	0.723
Random forest	0.928	0.837	0.881	0.858	0.983	0.949
XGBoost	0.927	0.821	0.901	0.859	0.983	0.951

**Fig. 1.** Precision, recall, and F1-score by model on the independent test set.

4 Results

4.1 Predictive Performance

Table 2 reports independent test performance. The two ensemble models clearly outperform the baselines. Logistic regression achieves an accuracy of 0.839, but recall is only 0.548. This is a practically important discrepancy: the model classifies many observations correctly overall while failing to identify almost half of the observed defaults. The decision tree improves recall to 0.834 and F1-score to 0.826, confirming that nonlinear structure is relevant.

Random forest obtains the highest accuracy (0.928) and precision (0.837). XGBoost sacrifices 0.016 precision relative to random forest but increases recall from 0.881 to 0.901. Its F1-score is marginally higher (0.859 versus 0.858), and it has the highest PR-AUC (0.951). The identical ROC-AUC values show that both ensembles rank observations very effectively, but the precision-recall balance at the evaluated threshold differs.

Figure 1 makes the operating trade-off visible. The result does not imply that XGBoost is always preferable. A lender facing a high cost for missed defaults may prefer its higher recall, whereas an institution seeking to reduce unnecessary reviews may favour the slightly higher precision of random forest. The important finding is that overall accuracy alone would not support this choice.

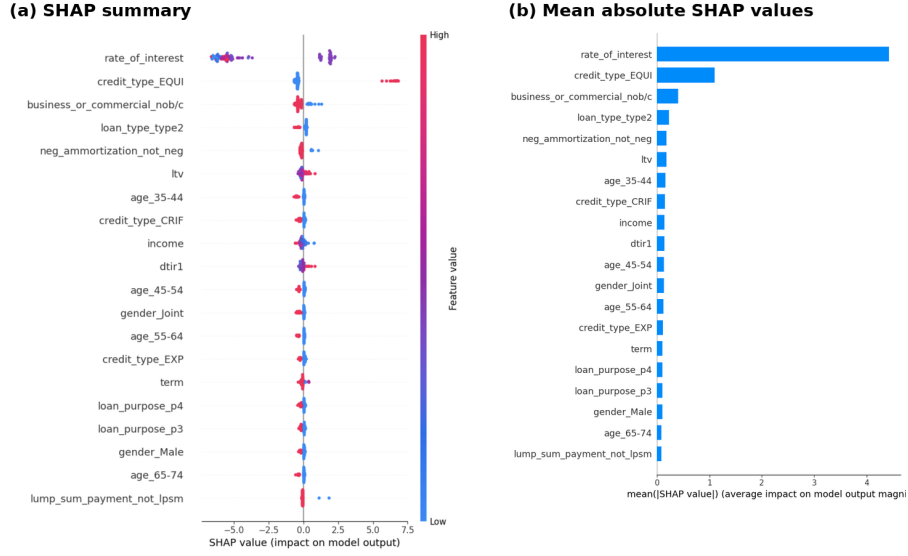


Fig. 2. Global explanation of the XGBoost model: (a) SHAP summary and (b) mean absolute SHAP importance.

4.2 Global Explanation of the XGBoost Model

Figure 2 presents two complementary global views. The SHAP summary plot combines feature rank, attribution direction, and feature value. The mean absolute SHAP plot aggregates magnitude and therefore provides a simpler importance ordering.

Interest rate is the dominant predictor by a substantial margin. Encoded credit-type indicators, business or commercial loan purpose, loan type, negative-amortisation status, age bands, loan-to-value ratio, and income also contribute. The presence of one-hot encoded categorical indicators requires careful interpretation. For example, the importance attached to a particular credit category is relative to the omitted reference category and the other encoded indicators, not an independent causal effect.

A SHAP dependence analysis suggests a nonlinear relationship between interest rate and the model output. Across the observed range, higher interest rates are generally associated with larger SHAP contributions toward default, with a marked change around approximately 4%. The colour encoding indicates interaction with the co-applicant credit-type indicator. This is an explanation of model behaviour, not evidence that changing the interest rate would cause the same change in default risk. Interest rate may reflect underwriting decisions and borrower characteristics that are also represented elsewhere in the data.

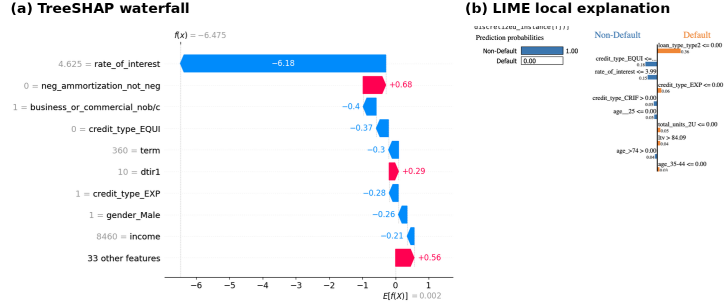


Fig. 3. Local explanations for the same selected test observation: (a) TreeSHAP waterfall and (b) LIME surrogate explanation.

4.3 Local Explanations

The selected test observation is predicted as non-default with high confidence. The TreeSHAP waterfall in Fig. 3(a) starts from the model’s expected output and shows how each feature moves the prediction. A relatively low interest rate is strongly protective for this case. Negative-amortisation status, business-purpose and credit-type indicators, term, age, gender, and income make smaller contributions in both directions. The final output is the net result of these effects rather than a single-variable rule.

The LIME explanation in Fig. 3(b) gives a more compact local narrative. It also identifies loan type, credit type, interest rate, loan-to-value ratio, and age-related indicators as influential. The broad agreement between the methods increases confidence that the main local story is not an artefact of one visualisation. However, the sign and magnitude of individual contributions are not directly comparable because SHAP decomposes the original model output while LIME fits a local surrogate to perturbed samples.

4.4 Case-Level Explanation Diagnostics

Table 3 summarises the explanation diagnostic. TreeSHAP requires approximately half the runtime of LIME in this implementation. Its stability score is 1.000 because repeated evaluation is deterministic for the fixed model, background configuration, and observation. LIME’s score of 0.551 indicates moderate variation across repeated perturbation-based explanations.

The fidelity result reflects the different constructions. TreeSHAP is an additive decomposition of the tree model, so the feature contributions reconstruct the selected model output up to numerical precision. LIME approximates the local decision surface with a sparse linear model and therefore attains lower fidelity. LIME’s advantage is compactness: six features account for approximately 80% of the displayed contribution compared with seven for SHAP. This can make the explanation easier to communicate, but compactness should not be confused with correctness.

Table 3. Case-level comparison of LIME and TreeSHAP.

Diagnostic	LIME	TreeSHAP
Runtime (seconds)	0.1160	0.0602
Normalised stability	0.5514	1.0000
Local fidelity	0.5876	$\approx$ 1.0000
Features for approximately 80% contribution	6	7

5 Discussion and Practical Lessons

5.1 Prediction and Explanation Form a Coupled Design Choice

The central application lesson is that the predictive model and the explanation method cannot be selected independently. XGBoost is attractive because it detects a larger proportion of defaults while maintaining strong precision and ranking performance. Once XGBoost is selected, TreeSHAP becomes a natural explanation component because it is computationally efficient for tree ensembles and exactly additive for the fitted model. This pairing is technically coherent.

LIME remains useful, but for a different purpose. Its sparse presentation can support communication with a reviewer, customer-service agent, or other non-specialist. The observed instability and lower fidelity mean that it should not be the sole explanation used for audit or model validation. A practical implementation can therefore use TreeSHAP as the primary analytical explanation and LIME as an optional communication layer, provided that repeated-run checks and clear disclaimers are included.

This distinction responds to a common weakness in applied XAI projects: an explanation is selected because the plot is familiar rather than because its operational properties match the governance objective. Explanation requirements should be specified in the same way as predictive requirements. An audit explanation prioritises reproducibility and faithfulness; a user-facing explanation prioritises clarity and brevity; a model-development explanation prioritises global patterns, interactions, and error analysis. No single plot satisfies all three purposes.

5.2 Accuracy Is Not a Sufficient Selection Metric

The logistic-regression result illustrates why accuracy is unsafe as the primary metric for imbalanced default prediction. An accuracy of 0.839 may initially appear acceptable, but recall of 0.548 reveals that many defaults remain undetected. The decision tree substantially improves sensitivity, and the ensembles improve both discrimination and minority-class performance.

Random forest and XGBoost are statistically close on several metrics. The operational decision therefore depends on the relative cost of false negatives and false positives. The present threshold favours XGBoost when missed defaults are considered more costly. In a production setting, the next step would be threshold

optimisation under explicit business costs, followed by probability calibration and stress testing. The current study does not perform these additional steps, so the reported model should be viewed as a strong candidate for decision support rather than a production-ready credit policy.

5.3 Feature Importance Is Not Causal Evidence

The global analysis provides plausible risk signals: interest rate, loan-to-value ratio, credit type, and loan characteristics are consistent with domain expectations. Plausibility is useful as a diagnostic because an implausible ranking may reveal leakage, coding errors, or unstable modelling. It does not establish causality. Interest rate may be both a contractual feature and a proxy for risk assessment already performed by a lender. Encoded categories can also divide one conceptual variable into multiple indicators, dispersing or concentrating attribution.

Correlated predictors are a particular concern. Standard SHAP values allocate contribution according to the explanation’s background assumptions, and different treatments of dependence can change the attribution assigned to correlated variables [1]. Consequently, the feature ranking should be interpreted as a description of the fitted model under a specific preprocessing and explanation configuration. It should not be used directly to justify policy changes without additional causal and domain analysis.

5.4 A Governance-Oriented Deployment Pattern

The results suggest a four-stage deployment pattern:

1. **Predictive validation:** evaluate discrimination, recall, precision, calibration, and threshold-specific costs on time-separated or external data.
2. **Explanation validation:** test faithfulness, stability, sensitivity to background data, and consistency across important borrower segments.
3. **Human review design:** translate model outputs into a review interface that distinguishes evidence, model attribution, and policy rules.
4. **Monitoring:** track data drift, class prevalence, performance by subgroup, explanation drift, and override outcomes.

The current experiment addresses the first two stages partially. It demonstrates strong predictive discrimination and provides an initial explanation audit. It does not test external validity, fairness, calibration, or longitudinal drift. Making these gaps explicit is preferable to presenting post-hoc plots as proof of responsible deployment.

6 Limitations and Threats to Validity

The study has several limitations. First, it uses one public dataset. Its large sample size supports stable within-dataset estimation, but the records may not

represent current portfolios, institutions, or jurisdictions. External validation is necessary before the findings can be generalised.

Second, the experiment compares four widely used models but does not include cost-sensitive learning, calibrated boosting, monotonic constraints, interpretable boosting machines, or modern neural architectures. The study therefore evaluates a practical baseline pipeline rather than the full model space. The choice is consistent with the application objective, but it limits claims about the globally best algorithm.

Third, SMOTE changes the training distribution. It improves the learner’s exposure to minority patterns, but synthetic examples may not correspond to feasible borrower profiles in every region of the feature space. Future work should compare SMOTE with class weighting and threshold adjustment, ideally within a fully nested cross-validation pipeline.

Fourth, the explanation comparison is diagnostic rather than comprehensive. The reported stability, fidelity, and runtime refer to a selected observation and the original implementation. A stronger evaluation would repeat the analysis across many correctly and incorrectly classified cases, across risk deciles, and across demographic and loan segments. It would also examine sensitivity to LIME parameters and SHAP background samples.

Fifth, the study does not evaluate fairness. Several encoded variables, including age and gender indicators, can be sensitive or can correlate with protected characteristics. An explanation that reveals their use is a warning signal, not evidence that their use is appropriate. A deployment study should assess subgroup performance, disparate error rates, proxy effects, and the legal and policy basis for each feature.

Finally, the model predicts associations in historical data. Neither SHAP nor LIME converts those associations into causal explanations. The explanations should be used to inspect model behaviour and support challenge, not to claim why a borrower will default in the real world.

7 Conclusion

This paper evaluated an integrated prediction-and-explanation workflow for imbalanced loan default classification. XGBoost achieved the best balance of recall, F1-score, and PR-AUC, while random forest delivered nearly identical overall discrimination and slightly higher precision and accuracy. The comparison shows that model selection depends on the operating objective: a small difference in accuracy can conceal a meaningful difference in the number of defaults detected.

The explanation analysis further shows that explanation methods have measurable operational properties. TreeSHAP produced a faithful, deterministic, and efficient decomposition of the selected XGBoost prediction. LIME generated a concise and intuitive local summary, but was less stable and less faithful in the case-level diagnostic. The two methods are therefore complementary rather than interchangeable.

The broader implication is that explainable AI should not be added after a classifier has been selected. Predictive performance, explanation reliability, communication needs, and governance controls should be designed together. The present pipeline offers a practical starting point, but external validation, calibration, threshold-cost analysis, fairness assessment, and explanation testing across many cases remain necessary before operational use.

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